

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python (Production), C++, SQL, Javascript

Deep Learning: Model Evaluation, Feature Engineering, TensorFlow, PyTorch

MLOps: Docker, Kubernetes, AWS SageMaker, Azure DevOps, CI/CD, FastAPI, Automated Testing (pytest), Databricks, Real-time System Optimization, Production Deployment

Data Engineering: PySpark, Airflow, ETL Pipelines, Data-Pipelines, Training-Pipelines, Data Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated robust deep learning models for metrological applications, focusing on operational stability and performance.
- Explored and implemented physics-informed hybrid loss functions to enhance model accuracy and ensure stable prediction capabilities for real-world applications.
- Collaborated closely with PhD researchers to prepare and validate models for publication, ensuring high code quality and adherence to research standards.
- Developed dashboards for comprehensive data visualization and real-time model performance monitoring, crucial for operational oversight.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Enhanced vitiligo segmentation accuracy by 94% and increased inference speed by 5x using advanced AI algorithms, deployed efficiently with Docker and FastAPI.
- Developed a real-time tennis foul detection system utilizing TensorRT for optimized inference, reducing compute cost and processing time by 6x in a production environment.
- Managed robust MLOps pipelines on AWS SageMaker, actively addressing YOLO model data drift and maintaining critical Kubernetes infrastructure for continuous operational stability.
- Contributed to CI/CD pipeline development, ensuring automated testing and reliable deployment of ML services into cloud environments.
- Implemented automated tests (e.g., pytest) for ML services, guaranteeing code quality and system reliability in production.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed backend services for AI-based applications using Python and Flask, ensuring robust functionality and maintainability for internal tools.

- Designed and integrated AI chatbot functionality via the OpenAI API, focusing on scalable and stable API interactions within a web service architecture.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built and maintained automated ETL pipelines for structured and unstructured data, ensuring reliable data ingestion for client analytics use cases.
- Automated data ingestion and cleaning workflows using Python, significantly streamlining data processing for downstream analytical models and operational use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, utilizing AlphaFold for feature extraction and taxonomic bias mitigation, demonstrating strong ML understanding.
- Increased model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces, focusing on robust and impactful features for evaluation.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system using advanced time-series models and data streams.
- Implemented data processing pipelines and prediction services, ensuring operational efficiency and timely delivery of forecasts.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed real-time signal processing system integrating temporal dependencies for music generation.
- Implemented transformer-based models for sequence prediction in audio generation tasks.

Hugging Face

Open-Source

Contributor

- Co-authored comprehensive technical documentation and implementation pages for Zero-shot Image Classification on the Hugging Face Hub, demonstrating strong technical communication and code quality focus.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling.
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor.
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python